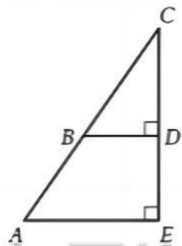


SAT UZBEKISTAN

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Module-1

1.



In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 56° , and $AE = 22(BD)$. What is the measure of angle CAE ?

- (A) 22°
- (B) 56°
- (C) $(22 + 56)^\circ$
- (D) $(22 \cdot 56)^\circ$

2.

A line segment that has a length of 125 centimeters (cm) is divided into three parts. One part is 49 cm long. The other two parts have lengths that are equal to each other. What is the length, in cm, of one of the other two parts of equal length?



3.

In the xy -plane, which of the following does NOT contain any points that are part of the solution set to $2x - 6y > 12$?

- (A) The region where $x < 0$ and $y > 0$
- (B) The region where $x < 0$ and $y < 0$
- (C) The region where $x > 0$ and $y > 0$
- (D) The x -axis

4.

In the xy -plane, line k and line ℓ are perpendicular and intersect at the point $(2, 9)$. If line k is defined by the equation $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on line ℓ ?

(A) $(3, 9 - \frac{1}{m})$

(B) $(3, 9 + \frac{1}{m})$

(C) $(3, 9 - m)$

(D) $(3, 9 + m)$

5.

The function f is defined by $f(x) = ab^{\frac{x}{n}}$, where a , b , and n are constants, and b and n are integers. If $f(5) = 7$ and $f(8) = 189$, what is the value of $f(10)$?



6.

The perimeter of an equilateral triangle is 876 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?



7.

In triangle ABC , the measure of angle A is 58° and $AC = 30$. In triangle PQR , the measure of angle P is 58° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

(A) $AB = 35$ and $PQ = 35$.

(B) $AB = 35$ and $QR = 140$.

(C) The measures of angle B and angle R are 34° and 88° , respectively.

(D) The measures of angle B and angle Q are 58° and 34° , respectively.

8.

The edge length, in inches, of cube Y is $\frac{3}{80}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n ?

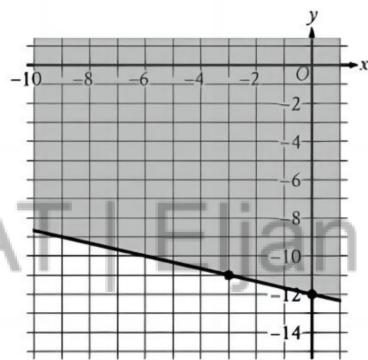
(A) $\frac{9}{7,300}$

(B) $\frac{27}{3,698}$

(C) $\frac{3}{80}$

(D) $\frac{9}{43}$

9.



The shaded region shown represents the solutions to $rx + ty \geq -36$, where r and t are constants. What is the value of $r + t$?

(A) 4

(B) 2

(C) -2

10.

For the linear function p , $p(c) = -2$, where c is a constant, $p(4) = 38$, and the slope of the graph of $y = p(x)$ in the xy -plane is 8. For the linear function t , $t(c) = -3$ and $t(5) = 51$. What is the slope of the graph of $y = t(x)$ in the xy -plane?

11.

To study fluctuations in leaf water potential, samples of wood were taken from 25 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 3.000 centimeters. This cube has a density of 0.220 grams per cubic centimeter. What is the mass of this cube, in grams?

12.

In 8 days, a polar bear ate 34.4 pounds of fat. Which equation describes the amount of fat y , in pounds, the polar bear ate in these 8 days?

(A) $y = 8$

(B) $y = \frac{34.4}{8}$

(C) $y = 34.4$

(D) $y = 34.4 + 8$

13.

A rectangle has a length that is 49 times its width. The function $y = (49w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $w > 0$. Which of the following is the best interpretation of $49w$ in this context?

- (A) The length of the rectangle, in feet
- (B) The area of the rectangle, in square feet
- (C) The difference between the length and the width of the rectangle, in feet
- (D) The width of the rectangle, in feet

14.

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 7)$ and passes through the points $(2, 53)$ and $(-1, 191)$. What is the value of $f(-2) - f(0)$?

- (A) 145
- (B) 237
- (C) 368
- (D) 421

15.

A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 24% every 22.96 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

- (A) $M = 100(0.24)^{t+22.96}$
- (B) $M = 100(0.24)^{\frac{t}{22.96}}$
- (C) $M = 100(0.76)^{t+22.96}$
- (D) $M = 100(0.76)^{\frac{t}{22.96}}$

16.

The difference between the measure of angle A and the measure of angle B is $-\frac{5}{12}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B , in degrees?

- (A) $-\frac{5}{12} \cdot \frac{180^\circ}{\pi}$
- (B) $-\frac{5}{12} \pi \cdot \frac{\pi}{180}$
- (C) $-\frac{5}{12} \pi \cdot \frac{180^\circ}{\pi}$
- (D) $-\frac{5}{12} \pi \cdot \frac{360^\circ}{\pi}$

17.

A school is ordering tablet computers for its students. The tablets cost \$170 each, and 7% tax is added to the total cost of the order. If the school can spend no more than 53,400 dollars on the tablets, including tax, what is the maximum number of tablets that the school can order?

18.

$$y = -0.5$$

$$y = x^2 + 8x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a ?

19.

A circle in the xy -plane has its center at $(-7, 3)$ and has a radius of 9. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

20.

A cooking school is offering a promotion where the first class is free, the second class is half off the regular price, and the remaining classes are regularly priced. If the regular price of a class is \$22.80, which function f gives the total cost, in dollars, of x classes taken using this promotion, where $x \geq 2$?

(A) $f(x) = 22.80(x - 1) + 11.40$

(B) $f(x) = 22.80(x - 2) + 11.40$

(C) $f(x) = 22.80(x - 1) + 11.40(x - 2)$

(D) $f(x) = 22.80(x - 2) + 11.40(x - 1)$

21.

$$3x = 45y - 27$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

(A) $x = 5y$

(B) $\frac{1}{3}x = 5y$

(C) $x = 15y - 9$

(D) $\frac{1}{3}x = 15y - 9$

22.

The function f is defined by $f(x) = -10(3)^x + \frac{1}{k}$, where k is a constant. If $y = f(x)$ is graphed in the xy -plane, what is the y -coordinate of the y -intercept of the graph?

(A) -10

(B) $\frac{1-10k}{k}$

(C) $\frac{1}{k}$

(D) $\frac{1-k}{k}$

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Module-2

1.

An object is launched from a height of 121 feet above the ground. A quadratic function f models the height above the ground, in feet, of the object t seconds after it is launched. According to the model, 2 seconds after the object is launched, it reaches a maximum height of 185 feet above the ground. Which equation defines f ?

(A) $f(t) = -16(t + 185)^2 - 2$

(B) $f(t) = -16(t - 185)^2 + 2$

(C) $f(t) = -16(t + 2)^2 - 185$

(D) $f(t) = -16(t - 2)^2 + 185$

2.

$$T(x) = 67 - 8x$$

The function T estimates the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), in a certain city one day in January, where x is the number of hours after 10 a.m. and $1 \leq x \leq 5$. According to the function, what was the estimated decrease in temperature each hour, in $^{\circ}\text{F}$, in this city during this time period?

(A) 4

(B) 8

(C) 59

(D) 67

3.

A rectangle has a length that is 92 times its width. The function $y = (92w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $w > 0$. Which of the following is the best interpretation of $92w$ in this context?

(A) The length of the rectangle, in feet

(B) The area of the rectangle, in square feet

(C) The difference between the length and the width of the rectangle, in feet

(D) The width of the rectangle, in feet

4.

The positive number a is 2,047% of the sum of the positive numbers b and c , and b is 89% of c . What percent of b is a ?

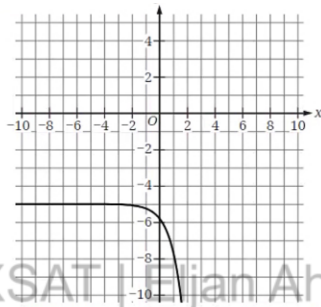
(A) 21.36%

(B) 38.69%

(C) 2,300%

(D) 4,347%

5.



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a , b , and c are constants. For how many values of x does $f(x) = 0$?

(A) Three

(B) Two

(C) One

(D) Zero

6.

A triangular prism has a height of 3 centimeters (cm) and a volume of 60 cm^3 . What is the area, in cm^2 , of the base of the prism? (The volume of a triangular prism is equal to Bh , where B is the area of the base and h is the height of the prism.)

7.

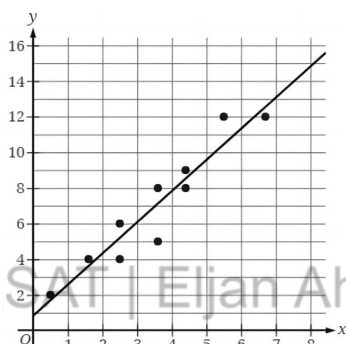
A school is ordering tablet computers for its students. The tablets cost \$164 each, and 6% tax is added to the total cost of the order. If the school can spend no more than 44,800 dollars on the tablets, including tax, what is the maximum number of tablets that the school can order?

8.

In one week in 2017, a technician earned a total of \$1,055 by working at her regular job and at a second job doing part-time work. The equation $20h + 17c = 1,055$ represents this situation, where h is the number of hours worked at her regular job and c is the number of hours worked at her second job. Which of the following is the best interpretation of 20 in this context?

- (A) The amount, in dollars, the technician earned for each hour she worked at her regular job
- (B) The amount, in dollars, the technician earned for each hour she worked at her second job
- (C) The number of hours the technician worked in one week at her regular job
- (D) The number of hours the technician worked in one week at her second job

9.



Which of the following equations best represents the line of best fit shown?

- (A) $y = 0.8 + 1.7x$
- (B) $y = 0.8 - 1.7x$
- (C) $y = -0.8 + 1.7x$
- (D) $y = -0.8 - 1.7x$

10.

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side lengths of 7 and 10, which inequality represents the possible lengths, x , of the third side of the triangle?

- (A) $x < 17$
- (B) $x > 17$
- (C) $3 < x < 17$
- (D) $x < 3$ or $x > 17$

11.

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 4)$ and passes through the points $(2, 27)$ and $(-1, 96)$. What is the value of $f(-2) - f(0)$?

- (A) 73
- (B) 119
- (C) 184
- (D) 211

12.

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 7.70 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

13.

The positive number a is 2,136% of the sum of the positive numbers b and c , and b is 89% of c . What percent of b is a ?

(A) 4,536%

(B) 2,400%

(C) 40.37%

(D) 22.25%

14.

$$f(x) = 5(g(x)) - 3$$

$$g(x) = |12x - 7|$$

The functions f and g are defined by the given equations. What is the value of $f(-10)$?

(A) -53

(B) 127

(C) 632

(D) 643

15.

For $x > 0$, the function f is defined as follows:

$f(x)$ equals 293% of x

Which of the following could describe this function?

(A) Decreasing exponential

(B) Decreasing linear

(C) Increasing exponential

(D) Increasing linear

16.

A certain elephant calf weighed 226 pounds when it was born. The elephant calf gained an average of 3 pounds per day in its first year of life. Which equation models this situation, where y is the estimated weight of the elephant calf, in pounds, x days after it was born?

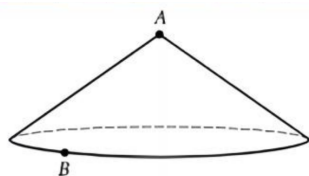
(A) $y = 226 + 3x$

(B) $y = 226 - 3x$

(C) $y = 3 + 226x$

(D) $y = 3 - 226x$

17.



Note: Figure not drawn to scale.

For the right circular cone shown, B is a point on the circumference of the base, and the length of segment AB (not shown) is 84 centimeters. If the height of the cone is 42 centimeters and the volume of the cone is $k\pi$ cubic centimeters, what is the value of k ?



18.

A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 23% every 11.69 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

(A) $M = 100(0.77)^{\frac{t}{11.69}}$

(B) $M = 100(0.77)^{t+11.69}$

(C) $M = 100(0.23)^{\frac{t}{11.69}}$

(D) $M = 100(0.23)^{t+11.69}$

19.

Harper has \$110 in an account. Each year she expects to have 2.7% more money in the account than she had the previous year. Which of the following models best describes how Harper expects the money in her account to change over time?

(A) Decreasing exponential

(B) Decreasing linear

(C) Increasing exponential

(D) Increasing linear

20.

A student needs at least 190 signatures from classmates to run for class president. The student collects 30 signatures on Monday and collects enough additional signatures on Tuesday to run for class president. Which inequality represents the number of additional signatures, s , that the student collects on Tuesday?

(A) $s + 30 \leq 190$

(B) $s + 30 \geq 190$

(C) $s - 30 \leq 190$

(D) $s - 30 \geq 190$

21.

The price of an object increased from \$16 to \$80. What was the percent increase in the price of the object?

(A) 20%

(B) 64%

(C) 80%

(D) 400%

22.

$$g(x) = x^3 + ax^2 + bx + c$$

The function g is defined by the given equation, where a , b , and c are integer constants. The zeros of the function are -2 , -7 , and 6 . What is the value of a ?

(A) -84

(B) -3

(C) 3

(D) 84